Purview CT-EDLP Egress-Control Target State Documentation

Summary

This target-state design defines 11 DLP policies and 11 matching rules, organized around distinct data-egress activities. It groups six device controls, three browser and cloud controls, one application control, and one communication control. Separating these channels makes it possible to validate, tune, and enable each control independently rather than applying one undifferentiated restriction to every workflow.

The purpose of this policy set is to reduce the risk of protected information leaving approved environments through removable storage, printing, clipboard operations, Bluetooth, remote sessions, network shares, web uploads, restricted applications, or external email. Sensitivity labels provide the intended classification-based starting point. The rules connect a content condition to an egress action, so protection can follow information across different ways of sharing or transferring it. No justified-override path is configured in the baseline.

All policies default to TestWithoutNotifications for silent simulation. This allows administrators to assess matches and business impact before enforcing blocks. Notification-enabled simulation can be used to validate user guidance, followed by staged enforcement after conditions, device coverage, destinations, and operational response have been verified. The attached script now builds the ten endpoint rules from the selected label GUIDs, which aligns them with the intended selected-label design. The Exchange rule (CT-EDLP-11) still uses a broader condition that matches any labeled content, so its effective scope must be distinguished from the intended selected-label design.

This set is an extensible egress-protection baseline. Sensitive information types (SITs) can be added to detect specific categories of information, such as personal identifiers or financial data, where the policy location supports them. Other supported conditions and exceptions can refine which content, users, recipients, destinations, or activities are covered. For example, an additional rule could protect content containing a chosen SIT even when it is not labeled; a combined condition could restrict a particular egress action only when both classification and sensitive content criteria are met. These are design options, not conditions already deployed by this baseline.

Extensions should retain the channel-specific structure and use separate rules or policies when workloads require different conditions or actions. Validate condition compatibility, match logic, SIT confidence and instance thresholds where applicable, exclusions, rule interactions, notifications, and expected handling of unlabeled content. Test every extension in simulation before enforcement; do not assume that adding a condition broadens coverage, since combining conditions with AND can narrow matches while OR can broaden them.

The following visual summarizes the intended channel-based protection model. Actual coverage depends on the configured conditions, supported workload capabilities, and successful validation.

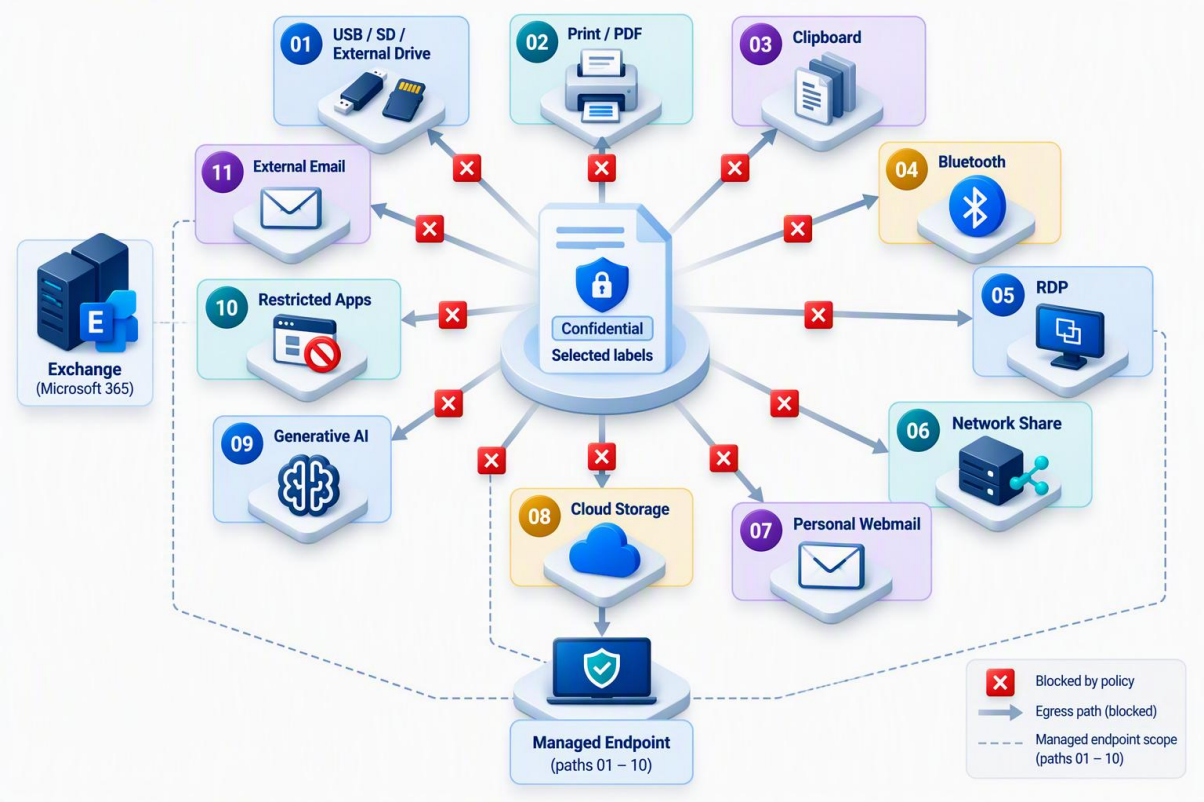


Figure 1: A selected-label document is surrounded by 11 channel-specific hard-block controls spanning managed endpoints and external Exchange email.

Common Control Outcome

All 11 rules share the following intended operational model:

- Trigger: Content carries one of the sensitivity labels selected for protection.
- Action: Block the defined egress activity.
- Override: No justified override is configured.
- User: Notify the last modifier when enforcement is active.
- Policy tip: Display common English and German guidance.
- Alert: Generate an alert for the configured incident mailbox.
- Incident: Generate an incident report for the same mailbox.
- Severity: High.
- Priority: 0 within the policy.
- Default mode: TestWithoutNotifications.

The common user message is:

- Blocked: content bears a sensitivity label that restricts this egress path. Contact IT Security.
- Blockiert: Vertraulichkeitsstufe schränkt diesen Übertragungsweg ein. IT-Sicherheit kontaktieren.

Incident reports include the title, service, matched item, matched rules, detections, document last modifier, sensitivity label, and severity.

Simulation Versus Enforcement

State	Block	User Tip	Operational Purpose
TestWithoutNotifications	No production enforcement	No	Collect silent match evidence.
TestWithNotifications	Simulated	Yes	Validate the user experience.
Enable	Yes	Yes	Apply the configured block and notification.

The design permits each policy mode to be managed independently. Silent simulation is the default for all 11 policies.

Detailed Policy and Rule Matrix

Policy	Rule	Group	Protected Trigger	Egress Action	Intended Result	Notification and Operations	Default Mode	Dependency / Readiness
CT-EDLP-01-Removable-Media	CT-EDLP-01-Removable-Media	Device	Selected labels	Copy to USB, SD card, or external drive	Block	Last modifier; bilingual tip; High alert and incident	Silent simulation	Managed Endpoint DLP devices
CT-EDLP-02-Print	CT-EDLP-02-Print	Device	Selected labels	Print or print to PDF	Block	Same common outcome	Silent simulation	Managed Endpoint DLP devices
CT-EDLP-03-Clipboard	CT-EDLP-03-Clipboard	Device	Selected labels	Clipboard copy or paste	Block	Same common outcome	Silent simulation	Managed Endpoint DLP devices
CT-EDLP-04-Bluetooth	CT-EDLP-04-Bluetooth	Device	Selected labels	Bluetooth transfer	Block	Same common outcome	Silent simulation	Windows endpoint and Bluetooth control validation
CT-EDLP-05-RDPCopy	CT-EDLP-05-RDPCopy	Device	Selected labels	RDP clipboard or drive redirection	Block	Same common outcome	Silent simulation	RDP activity validation
CT-EDLP-06-NetworkShare	CT-EDLP-06-NetworkShare	Device	Selected labels	Copy to SMB/network share	Block	Same common outcome	Silent simulation	Network-share impact validation

Policy	Rule	Group	Protected Trigger	Egress Action	Intended Result	Notification and Operations	Default Mode	Dependency / Readiness
CT-EDLP-07-WebmailUpload	CT-EDLP-07-WebmailUpload	Browser & Cloud	Selected labels	Upload to personal webmail	Block	Same common outcome	Silent simulation	Webmail service domain groups (Webmail - Global & Privacy, Webmail - Regional)
CT-EDLP-08-CloudStorage	CT-EDLP-08-CloudStorage	Browser & Cloud	Selected labels	Upload to unsanctioned storage	Block	Same common outcome	Silent simulation	Storage service domain groups (Cloud Storage - Global, Cloud Storage - Regional & Hosting)
CT-EDLP-09-GenerativeAI	CT-EDLP-09-GenerativeAI	Browser & Cloud	Selected labels	Paste or upload to AI websites	Future hard block	Same common outcome	Silent simulation	AI Hub, DLP for AI Apps, Generative AI Websites group, feature validation
CT-EDLP-10-UnallowedApps	CT-EDLP-10-UnallowedApps	Application	Selected labels	Access by restricted applications	Block	Same common outcome	Silent simulation	Restricted App Groups
CT-EDLP-11-Exchange	CT-EDLP-11-Exchange	Communication	Any label and external recipient (selected-label targeting pending)	External Exchange email	Block access for all matching external transactions	Same common outcome	Silent simulation	External-scope and label validation

The exact rule actions, notification recipients, alerting, reporting, severity, and scope are defined consistently across the 11 controls.

Device Egress

CT-EDLP-01 – Removable Media

CT-EDLP-01-RemovableMedia protects selected-label files against physical-media exfiltration. The rule blocks copying to:

- USB drives
- SD cards
- External hard drives
- Other removable-media destinations covered by the endpoint restriction

The outcome prevents a protected file from being moved from a managed endpoint to portable storage. The rule notifies the last modifier and produces a High-severity alert and incident report when enabled.

CT-EDLP-02 – Print

CT-EDLP-02-Print blocks printing of selected-label documents.

The control covers physical printing and print-to-PDF. This closes both paper-based and file-based print egress paths. The control applies across all configured Endpoint DLP locations.

CT-EDLP-03 – Clipboard Copy

CT-EDLP-03-Clipboard blocks clipboard operations involving selected-label content.

The outcome prevents protected information from being copied and pasted into unauthorized applications or other documents. It addresses content movement even when the original file itself is not transferred.

CT-EDLP-04 – Bluetooth Transfer

CT-EDLP-04-Bluetooth blocks selected-label file transfers through Bluetooth mechanisms.

The target outcome covers Bluetooth file-transfer utilities and operating-system-level Bluetooth sharing on managed Windows endpoints.

CT-EDLP-05 – RDP Copy

CT-EDLP-05-RDPCopy blocks protected content from leaving through Remote Desktop sessions.

The rule covers clipboard and drive-redirection activity. It prevents selected-label content from being transferred from the managed environment to an RDP-connected client.

CT-EDLP-06 – Network Share Copy

CT-EDLP-06-NetworkShare blocks copying selected-label files to SMB and network-share destinations.

The outcome restricts lateral movement to unmanaged or non-compliant file-server locations. It is distinct from removable-media protection because the destination is network-based rather than physically attached.

Browser and Cloud Egress

CT-EDLP-07 – Webmail Upload

CT-EDLP-07-WebmailUpload blocks browser uploads of selected-label files to personal webmail.

Sensitive service domain groups identify the websites and services to which Endpoint DLP applies configured upload and paste restrictions. For CT-EDLP-07-WebmailUpload, the deployment binds the CloudEgress and PasteToBrowser block actions to the tenant groups Webmail - Global & Privacy and Webmail - Regional, which are created beforehand by the tenant group import process and cover personal webmail services such as Gmail, Outlook.com, and Yahoo Mail. If neither group exists when the policy is deployed, the rule falls back to an unscoped CloudEgress and PasteToBrowser block and a configuration follow-up is recorded. The purpose is to prevent protected files from being uploaded as attachments to personal email accounts outside approved business communication channels. The policy name alone does not identify or configure those destinations.

Maintain a webmail destination inventory that records which services are restricted and which business services remain approved. Verify the domains used during attachment uploads rather than if a service's main website address covers every upload endpoint. Because CT-EDLP-07-WebmailUpload and CT-EDLP-08-CloudStorage reference separate service domain groups, webmail and storage destinations can be approved, tested, and maintained independently, provided the groups exist and do not overlap. Before enforcement, test protected-file uploads to restricted personal webmail and approved business destinations in supported browsers, and confirm the expected matches, user guidance, and actions. This endpoint webmail-upload control is distinct from CT-EDLP-11-Exchange, which addresses external email sent through Exchange.

CT-EDLP-08 – Cloud Storage Upload

CT-EDLP-08-CloudStorage blocks selected-label files from being uploaded to unsanctioned cloud storage.

Sensitive service domain groups identify the websites and services to which Endpoint DLP applies configured upload and paste restrictions. For CT-EDLP-08-CloudStorage, the deployment binds the CloudEgress and PasteToBrowser block actions to the tenant groups Cloud Storage - Global and Cloud Storage - Regional & Hosting, which are created beforehand by the tenant group import process. If neither group exists, the rule falls back to an unscoped block and a configuration follow-up is recorded. The policy name alone does not identify or configure those destinations. Storage services, such as Dropbox, Box, and personal Google Drive, represent a different business use case from personal webmail services, such as Gmail or Yahoo Mail: storage uploads place files in a repository, whereas webmail uploads typically attach files to messages.

Maintain separate destination inventories for storage and webmail so that ownership, approval decisions, testing, and any future exceptions can be managed independently. The two policies now reference separate service domain groups, but the separation is only effective when those groups exist, are maintained, and do not share domains. Confirm how the configured groups and rule actions interact in the tenant, including any overlap between CT-EDLP-07-WebmailUpload and CT-EDLP-08-CloudStorage. Paste-to-browser blocking does not trigger on label-matched content, because the endpoint paste control evaluates the pasted text independently of the source file's label; blocking paste to storage sites therefore requires a separate rule based on sensitive information types (SITs). Before enforcement, test uploads to both restricted and approved destinations in supported browsers and verify that the intended content conditions, notifications, and actions are applied without disrupting approved business workflows.

Why CT-EDLP-07 and CT-EDLP-08 Are Separate

Both controls use cloud-egress enforcement. They govern different destination classes.

- CT-EDLP-07 governs personal communication destinations.
- CT-EDLP-08 governs personal or unsanctioned storage destinations.

The separation supports different destination inventories, service domain groups, ownership, validation, and future exceptions. It is intentional channel specialization, not duplication.

CT-EDLP-09 – Generative AI Websites

CT-EDLP-09-GenerativeAI is a readiness-dependent target control.

Its intended outcome is to block pasting or uploading selected-label content to generative AI websites. Examples include ChatGPT, Gemini, and Copilot.com.

Enforcement requires validation of:

- Microsoft Purview AI Hub availability
- DLP for AI Apps licensing
- The AI website restriction in the target tenant
- Expected paste and upload event coverage

The deployment combines the dedicated AI restriction (AccessByAIAgent) with CloudEgress and PasteToBrowser blocks scoped to the built-in Generative AI Websites service domain group, which uses the same well-known identifier in every tenant. If the dedicated capability cannot be used, the domain-group restrictions remain the fallback design. This control should not move to enforcement until the capability and license are confirmed.

Application Egress

CT-EDLP-10 – Unallowed Applications

CT-EDLP-10-UnallowedApps prevents restricted desktop applications from accessing selected-label files.

When enabled, applications in Restricted App Groups cannot open, read, or copy protected content. This reduces exposure through unmanaged, unauthorized, or otherwise restricted software.

The control is not effective until Restricted App Groups are populated and validated. Application ownership and business-impact testing are therefore required before enforcement. Restricted apps and app groups are configured in the Microsoft Purview portal under Data loss prevention → Overview → Data loss prevention settings → Endpoint settings → Restricted apps and app groups.

Communication Egress

CT-EDLP-11 – Exchange Email (External)

CT-EDLP-11-Exchange blocks selected-label content from being sent through Exchange to recipients outside the organization.

The recipient scope is NotInOrganization. Internal Exchange email is unaffected. For matching external transactions, BlockAccessScope is set to All.

When enabled, the rule:

- Blocks the external email transaction.
- Notifies the last modifier.
- Shows the bilingual egress message.
- Generates a High-severity alert.
- Sends an incident report to the configured mailbox.

This is the recommended first enforcement candidate after simulation.

How the 11 Controls Add Up

The portfolio forms four complementary protection layers.

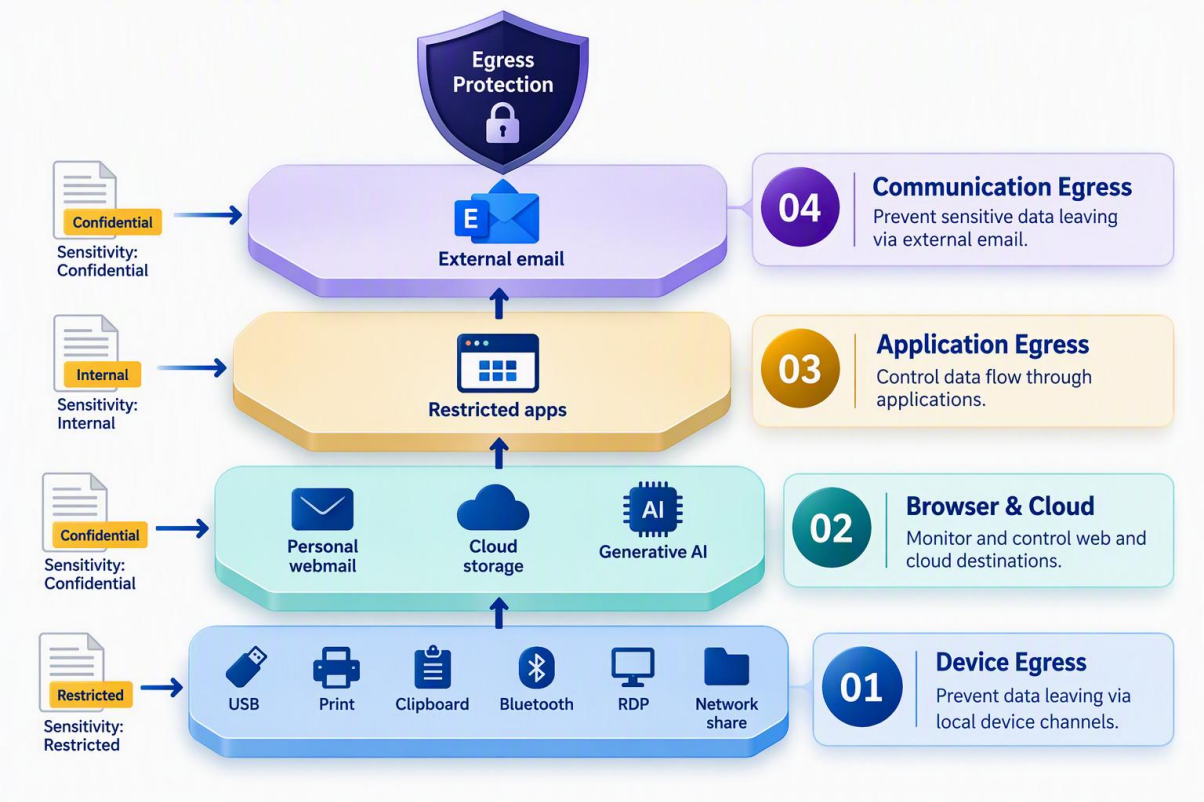


Figure 2: The 11 controls form four complementary layers: six device controls, three browser and cloud controls, one application control, and one external-email control.

Protection Layer	Controls	Combined Outcome
Device Egress	01–06	Stops physical transfer, printing, clipboard movement, Bluetooth sharing, RDP redirection, and network-share copying.
Browser & Cloud Egress	07–09	Stops uploads to personal webmail, unsanctioned storage, and validated generative AI destinations.
Application Egress	10	Stops restricted applications from accessing protected files.
Communication Egress	11	Stops external Exchange email containing protected content.

Together, these layers protect selected-label content across 11 distinct egress paths. Detection follows the content. Enforcement is selected according to the attempted action.

Defense in Depth and Overlap

The controls can match the same file at different moments. This is expected.

Clipboard Versus RDP

CT-EDLP-03 governs ordinary clipboard movement. CT-EDLP-05 governs clipboard and drive redirection inside an RDP session. The context differs even where clipboard behavior overlaps.

Removable Media Versus Network Shares

CT-EDLP-01 governs physically attached storage. CT-EDLP-06 governs SMB and network destinations. Both prevent file copying, but the destination type and risk path differ.

Webmail Versus Cloud Storage

CT-EDLP-07 governs personal mail services. CT-EDLP-08 governs unsanctioned storage. Both use cloud-egress controls, but they are scoped to different service domain groups and their domain classifications are intentionally different.

AI Websites Versus Clipboard and Browser Upload

CT-EDLP-09 focuses on AI destinations. A protected item could also encounter clipboard or browser controls before it reaches the AI service. This provides action-specific defense in depth.

Readiness and Validation

The intended selected-label outcome must be separated from current technical readiness. The endpoint rules now target the selected labels; the Exchange rule and the service domain groups still require validation.

Required Before Enforcement

1. Managed endpoints
    - Confirm all intended devices are available to Endpoint DLP.
    - Verify activity visibility across the target endpoint population.
2. Selected-label validation
    - Confirm the approved trigger-label list.
    - Verify that expected labels appear in Activity Explorer matches.
3. Specific-label condition validation
    - Confirm each rule matches only the selected labels.
    - The Exchange rule (CT-EDLP-11) still uses a condition that may evaluate any labeled content; the endpoint rules use the selected label GUIDs.

        1. This broader behavior must not be treated as the approved enforcement state.
4. Sensitive service domain groups
    - Confirm the Webmail - Global & Privacy and Webmail - Regional groups exist for CT-EDLP-07.
        1. Confirm the Cloud Storage - Global and Cloud Storage - Regional & Hosting groups exist for CT-EDLP-08.
5. Generative AI readiness
    - Validate AI Hub.
    - Validate DLP for AI Apps licensing.
    - Validate AI-site activity.
        1. Confirm the Generative AI Websites group is applied; use domain classification only as an approved fallback.
6. Restricted App Groups
        1. Define and test applications before enabling CT-EDLP-10.
7. Operational response
    - Monitor the incident mailbox.
        1. Confirm alerts and incident reports contain the expected service, matched item, detections, label, modifier, and severity.

Policy Functionality Test Cases

Use the following proposed test cases to verify each policy and rule before production enforcement. Run tests with authorized test accounts, onboarded test devices, synthetic files, and controlled destinations. Do not use real personal information, credentials, or production-sensitive data. Record the actual policy mode and effective rule configuration before every test.

Test Prerequisites and Expected Behavior

- Prepare equivalent files with an intended protected label, a non-target label, and no label. Record each file's name, type, label, and content.
- Check the effective content conditions first. The endpoint rules created by the attached script match the selected labels, so a non-target labeled file should not match CT-EDLP-01 to CT-EDLP-10. The Exchange rule still matches any labeled content; a non-target labeled file may therefore match CT-EDLP-11. If selected-label-only targeting is required there, treat that match as a configuration gap, not a successful acceptance result.
- Verify the policy and rule exist, include the test account and device or workload, and have reached the test environment. Check domain lists, restricted applications, and feature prerequisites where relevant.
- In TestWithoutNotifications, verify matching telemetry without expecting a production block or user tip. In TestWithNotifications, verify simulation evidence and supported user guidance without expecting enforcement. Use Enable only in an isolated, approved pilot to verify actual blocking.

- For every channel, attempt the same activity with a file that does not meet the effective content conditions. Expect no restriction from the tested rule; other policies may still apply.

Channel-Specific Functionality Tests

Policy and Matching Rule	Test Activity	Expected Enforcement Result
CT-EDLP-01-Removable-Media	Copy a matching file to an approved test USB drive; repeat with an SD card or external drive where available.	The removable-media copy is blocked in Enable mode. Verify the destination contains no usable copied file.
CT-EDLP-02-Print	Print a matching document to a test physical printer and separately to a PDF printer.	Printing is blocked in Enable mode for supported print paths. Verify no print job or usable output is produced.
CT-EDLP-03-Clipboard	Copy content from a matching file and attempt to paste it into a controlled destination application.	The configured clipboard-copy activity is blocked in Enable mode. Record source and destination applications and whether content reaches the destination.
CT-EDLP-04-Bluetooth	Attempt to send a matching file to an authorized test Bluetooth receiver using a supported transfer application.	The configured Bluetooth transfer is blocked in Enable mode. Verify the receiver does not obtain the protected file; validate platform and application coverage.
CT-EDLP-05-RDPCopy	In a controlled RDP session, attempt to copy a matching file through redirected drives; separately test clipboard redirection.	The configured RDP copy restriction blocks supported transfers in Enable mode. Record transfer direction and validate clipboard and drive behavior separately rather than assuming identical coverage.
CT-EDLP-06-NetworkShare	Copy a matching file to a controlled SMB share; separately test any approved share exception.	The configured network-share copy is blocked in Enable mode unless an effective exception applies. Verify the share does not contain a usable copy.
CT-EDLP-07-Web-mailUpload	Upload a matching file as an attachment to a restricted personal-webmail test account in a supported browser; repeat against an approved business destination.	The restricted upload is blocked in Enable mode when the domain configuration covers the upload endpoint. The approved-destination result must match the effective allow/block configuration.

Policy and Matching Rule	Test Activity	Expected Enforcement Result
CT-EDLP-08-CloudStorage	Upload a matching file to a restricted storage test account; repeat against approved storage and check overlap with the webmail policy.	The restricted upload is blocked in Enable mode. Approved-storage behavior follows the effective domain groups and exceptions; identify all rules that match. Do not expect label-based paste blocking; paste control requires a separate SIT-based rule.
CT-EDLP-09-GenerativeAI	Using synthetic content, separately test file upload and paste into an authorized AI test destination. Verify the deployed AI restriction and prerequisites first.	Verify blocking only for activities supported by the effective AI restriction in Enable mode. Record upload and paste outcomes separately. Unsupported coverage or failed rule creation is a readiness gap.
CT-EDLP-10-UnallowedApps	Open a matching file with an application configured as restricted; repeat with an approved application and a nonmatching file.	The configured restricted-app access is blocked in Enable mode. Approved-application and nonmatching-file results must follow the effective rule settings; the application itself need not be prevented from running.
CT-EDLP-11-Exchange	Send matching test content through Exchange to an external test recipient; separately send to an internal recipient and test a message with both recipient types.	Matching external email is restricted in Enable mode. Internal-only email should not match this rule's external scope. Verify actual delivery and mixed-recipient handling rather than assuming partial delivery.

Evidence and Acceptance Checks

For each case, record the policy and rule name, test account, device and operating system, application or browser, destination, file label, policy mode, test time, expected result, actual result, and evidence reference. Correlate Activity Explorer events with the attempted activity and identify any additional rules that influenced the outcome.

- Verify the intended action, user notification, and policy-tip wording in the applicable mode. In enforcement tests, confirm that no justified override is offered where the baseline defines a hard block.
- Check configured alerts and incident reports, including the matched rule, item, service, label, modifier, and severity. Allow for processing delays and aggregation; do not assume one separate alert or email per attempt.
- Mark each case Pass, Fail, or Not Tested. Use Not Tested for unsupported or unavailable channels and document the coverage gap explicitly.

- Accept a policy only after matching and nonmatching tests succeed, required business workflows remain available, and all observed behavior is explained by the effective configuration. Correct failures and repeat the affected tests before broader enforcement.
- If SITs or other conditions are added, repeat channel tests with synthetic matching and nonmatching content, including labeled and unlabeled files. Test AND/OR logic, confidence levels and instance thresholds where applicable, and exceptions.

Recommended Rollout

The rollout moves from evidence to controlled enforcement. The source requires an Activity Explorer review period of at least two weeks.



Figure 3: The rollout starts with at least two weeks of silent simulation, validates every dependency, enforces external Exchange email first, and then advances endpoint controls in stages.

The required sequence is:

1. Run all policies in silent simulation.
2. Review Activity Explorer for at least two weeks.
3. Confirm expected selected-label matches.
4. Review incident-mailbox events.
5. Resolve service domain group, AI, and Restricted App Group dependencies.
6. Validate specific-label targeting.
7. Enable CT-EDLP-11-Exchange first.
8. Stage endpoint controls by channel.

9. Monitor blocked workflows and alert quality after each enforcement step.

It covers local devices, operating-system actions, remote sessions, network destinations, browser destinations, cloud services, generative AI interactions, desktop applications, and external email. Every blocked event creates a user-facing outcome and a High-severity operational record. No justified-override path is part of the target design.